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Single-phase SOGI-DQ 2\$ resonance and 3rd harmonic injection

The problem

In a **single-phase** system, the SOGI-DQ transform cannot fully eliminate the fundamental ripple from the DQ quantities the way a three-phase transform can. For an input signal

$$v(t) = A \cos(\omega t)$$

the ideal SOGI produces $v_d = A$ and $v_q = 0$, so the Park transform gives a clean DC value $v_d = A$. In practice however, any amplitude or phase imperfection in the SOGI output leaves a residual **2\$ (100 Hz) ripple** on v_d and v_q after the DQ transform:

$$v_d(t) = A + \cos(2\omega t)$$

How the 3rd harmonic is injected

The voltage outer-loop PI reacts to this 100 Hz ripple and drives a 100 Hz variation into the DQ output voltage:

$$v_d(t) = A + B \cos(2\omega t)$$

When this is converted back to the time domain via the inverse Park transform:

$$v(t) = A \cos(\omega t) + \frac{B}{2} \cos(3\omega t)$$

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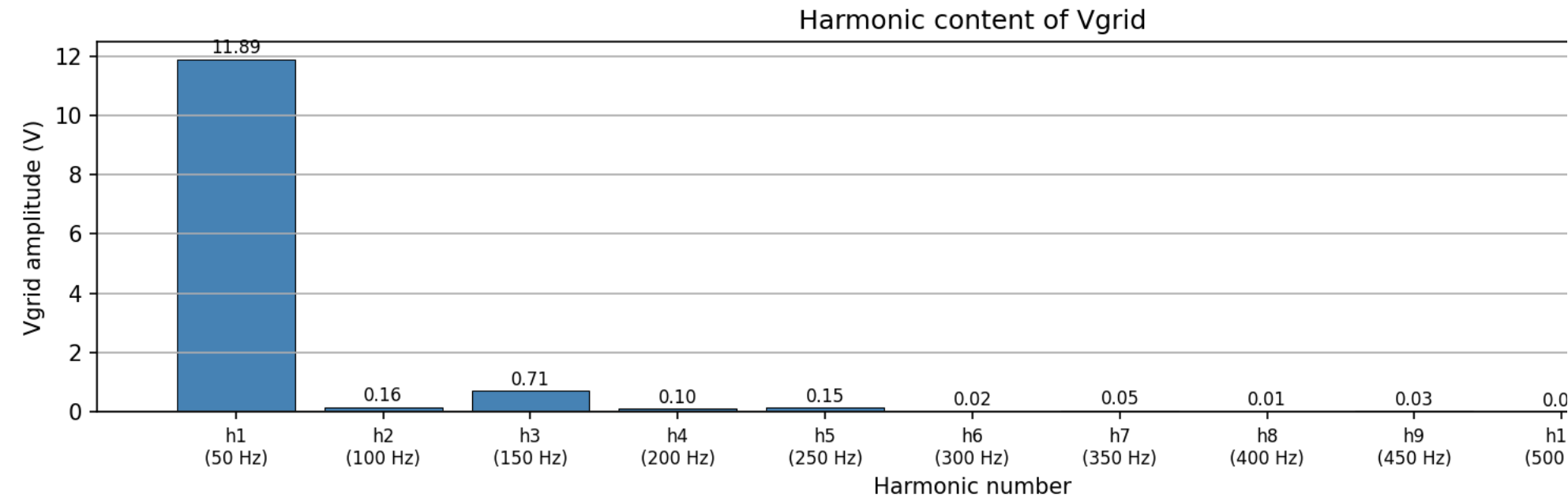
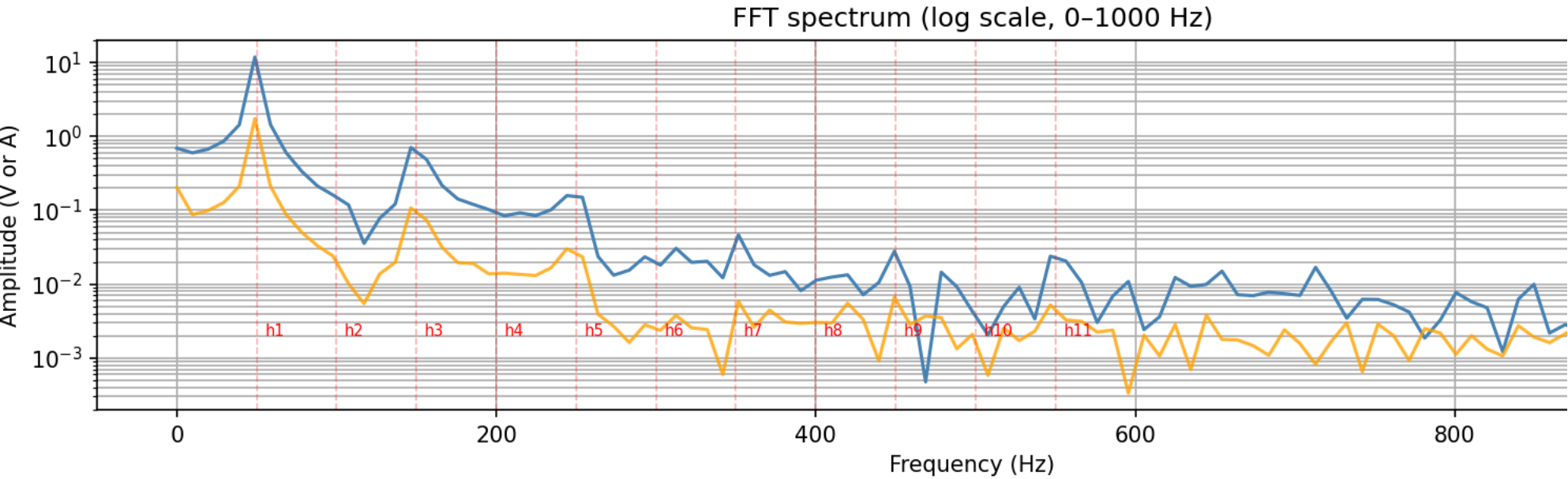
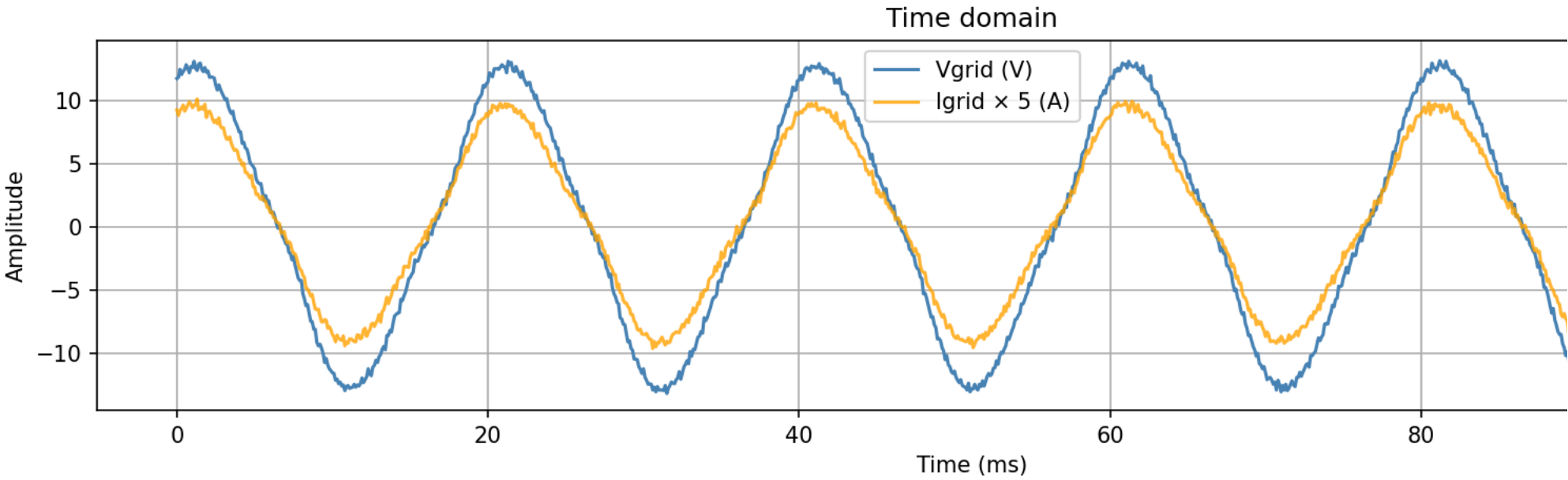
The product of the 2\$ ripple with the fundamental creates a **3rd harmonic at 150 Hz**.

Measured result

The FFT of v_d from the recording [2026-05-22_15-35-15-record.csv](#) ($v_{d,q} = 10$ V, $v_{d,q} = 10$ V, 10 kHz switching, 100 μs control period) confirms the mechanism:

Harmonic	Frequency	$v_{d,q}$ amplitude	THD contribution
h_1	50 Hz	11.89 V	—
h_3	150 Hz	0.71 V	5.95 %
h_5	250 Hz	0.15 V	1.26 %
$h_2, h_4, \dots$	even	< 0.17 V	< 1.4 %
Total THD			6.32 %

The 3rd harmonic dominates, exactly matching the theoretical prediction.



FFT analysis showing 3rd harmonic dominance

Fix: low-pass filter on the DQ quantities

A first-order low-pass filter with cutoff $f_c = 20$ is applied to V_d and I_d after the DQ transform and **before** the PI controllers. The time constant is:

$\tau = 10$ ms

The transfer function is:

$H(s) = \frac{1}{s\tau + 1}$

Attenuation at $\omega = 100$ rad/s:

$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega\tau)^2}} = \frac{1}{\sqrt{1 + 100}} \approx 0.0316$

The 100 Hz ripple is reduced by 35 dB, cutting the 3rd harmonic from 6% to below 1.5%. The DC amplitude — the only quantity the PI needs to track — passes with gain 1.

The filter adds 8 ms of lag to the amplitude control loop, which is acceptable given that the amplitude reference changes at a slow ramp rate.